

Billy SIEVERS

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Education

Ph.D. in Physics , <i>McMaster University</i> GPA: 3.57 / 4.00 • Theoretical Physics Group • NSERC PGS-D Recipient <i>Notable courses</i> : Mathematical and Computational Finance, Advanced Parallel and High Performance Computing.	Hamilton, ON	2023 – 2027
M.Sc. in Physics , <i>Memorial University of Newfoundland & Labrador</i> GPA: 4.00 / 4.00 • Gravity Group • Fellow of the School of Graduate Studies	St. John's, NL	2020 – 2023
B.Sc. Honours in Physics , <i>University of Calgary</i> GPA: 3.60 / 4.00 • International Student Scholarship Recipient	Calgary, AB	2016 – 2020

Technical Skills

Python (NumPy, SciPy, Matplotlib), C, Mathematica; Monte Carlo simulation, machine learning, spectral analysis; GPU computing (OpenACC), Linux, Git, $\text{\LaTeX}$; AI-assisted development.

Experience

Researcher , <i>Theoretical Physics Group, McMaster University</i> • Analyzed and delivered numerical solutions to ODEs that model black hole boundaries; most impactful contribution was a machine-learning component that sped up data collection by 50x. Programmed in Python and C. • Sought improvements by implementing GPU-parallelization methods to further optimize computation efficiency while regularly assessing performance and payoff. • Validated and challenged submissions as a referee for the academic journals ‘Classical & Quantum Gravity’ and ‘Physics Letters B’, rapidly understanding the models proposed in the articles and responding within deadlines. • Active contributor to the gravity research community through four peer-reviewed publications, with two more in preparation. Disseminated research results at national and international conferences multiple times a year.	Hamilton, ON	2023 – present
Teaching Assistant , <i>McMaster University</i>	Hamilton, ON	2023 – present
Teaching Assistant , <i>Memorial University of Newfoundland</i>	St. John's, NL	2020 – 2022
Founder and Sole Proprietor , <i>Billy's Squash Shop</i>	Hamilton, ON	2024 – present
Team-Lead, Instructor, and 1-on-1 Tutor , <i>MathPro Learning Centre</i>	Calgary, AB	2017 – 2023

Leadership and Extra-curricular Activities

Player & Equipment Manager , <i>McMaster Marauders Squash Team</i>	2025 – present
Chair / President , <i>McMaster Astronomy & Physics Graduate Student Association (MAPSA)</i>	2025 – present
Representative for Theoretical Physicists , <i>MAPSA</i>	2023 – present
Volunteer Telescope Operator , <i>Rothney Astrophysical Observatory, U of Calgary</i>	2016 – 2020
Vice-President , <i>Physics and Astronomy Students Association, U of Calgary</i>	2017 – 2018

Chosen Publications

- **K. T. B. Sievers**, L. Newhook, S. Muth, I. Booth, R. A. Hennigar, H. K. Kunduri, “Marginally Outer Trapped Tori in Black Hole Spacetimes”, *Phys. Rev. D* 109, 124023 (2024).
- R. A. Hennigar, H. K. Kunduri, **K. T. B. Sievers**, Y. Wang, “Spectrum of the Laplacian on the Page metric”, *J. Phys. A* 58, 405204 (2025).

Awards

NSERC Postgraduate Scholarship - Doctoral (\$101 000) • NSERC CRTAS - Research Travel Supplement (\$9 000)
McMaster GSA - Therese Quigley Award of Excellence for Graduate Student Leadership in Athletics (\$200)
Canadian Student & Postdoc Conference on Gravity - Best M.Sc. Presentation
University of Calgary - International Student Scholarship (\$60 000)